

Sampling Distributions

STAT 24510-30040

(Winter 2025)

Test statistics are functions of observed outcomes of random variables, they are crucial in statistical inference.

The most common normal-theory based test statistics are the χ^2 statistic, the t statistic, and the F statistic.

In this section, we study these common statistics and derive their probability distributions.

1 The χ^2 distribution

Motivation of χ^2 distribution

Population variance σ^2 is often estimated from a sample $X_1, \dots, X_n$ by an estimator of the form

$$c \sum_{i=1}^n (X_i - \bar{X})^2$$

where the choice of c depends on the criterion. For example, $c = \frac{1}{n-1}$ if the estimator is the unbiased sample variance (denoted as S^2 or s^2), $c = \frac{1}{n}$ if the estimator is the MLE, etc.

Probability distribution of the variance estimators are closely related to the chi-square distribution.

Chi-square distribution is denoted as χ^2 or χ_k^2 , where k is the degrees of freedom.

Definition of χ^2 distribution

If $Z_1, \dots, Z_n$ are independent standard normal random variables, then $Z_1^2 + \dots + Z_n^2$ is of χ^2 distribution with degrees of freedom n , that is,

$$Z_1^2 + \dots + Z_n^2 \sim \chi_n^2, \quad \text{where} \quad Z_1, \dots, Z_n \text{ i.i.d.} \sim N(0, 1)$$

Density of χ^2 distribution

The density function of the χ_n^2 distribution has the form

$$f_{\chi_n^2}(y) = \frac{1}{\Gamma(\frac{n}{2})} \left(\frac{1}{2}\right)^{\frac{n}{2}} y^{\frac{n}{2}-1} e^{-\frac{1}{2}y}, \quad y \geq 0.$$

Proof. We derive the χ^2 density function by mathematical induction on the degrees of freedom parameter n .

For $n = 1$, let

$$Y = g(Z_1) = Z_1^2$$

By the variable transformation formula, the probability density function of $Y = g(Z)$ has the form

$$f_Y(y) = \sum_{\{z: g(z)=y\}} f_Z(z) \left| \frac{1}{g'(z)} \right|$$

where the summation is over all z such that $g(z) = y$, and $f_Z(z)$ is the density function of Z .

Here $g'(z) = 2z$, and $Z = Z_1$ is of standard normal,

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

and $Y = Z_1^2$ has density function

$$f_Y(y) = \sum_{\{z: z^2=y\}} f_Z(z) \left| \frac{1}{g'(z)} \right| = \sum_{\{z: z=\pm\sqrt{y}\}} \frac{f_Z(z)}{|2z|} = \frac{f_Z(\sqrt{y})}{2|\sqrt{y}|} + \frac{f_Z(-\sqrt{y})}{2|-\sqrt{y}|} = \frac{1}{\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}}, \quad y \geq 0.$$

This is the density function of χ_1^2 distribution.

Recall that the density functions of the Gamma distribution with shape parameter α and rate parameter β is defined as

$$f(x) = \frac{1}{\Gamma(\alpha)} \beta^\alpha x^{\alpha-1} e^{-\beta x}, \quad x \geq 0. \quad (\alpha, \beta \geq 0)$$

where

$$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$$

is called the Gamma function, with the characteristic property

$$\Gamma(\alpha + 1) = \alpha \Gamma(\alpha), \quad \alpha > 0.$$

In particular,

$$\Gamma(k + 1) = k! \quad \text{for integer } k.$$

Using integration-by-parts formula, the mean and variance of $\text{Gamma}(\alpha, \beta)$ can be derived as

$$\mathbb{E}(X) = \frac{\alpha}{\beta}, \quad \text{Var}(X) = \frac{\alpha}{\beta^2}$$

Note that sometimes, the scale parameter $\theta = 1/\beta$ is used instead of β in Gamma distribution.

We can re-express the density function of χ_1^2 as

$$f_Y(y) = \frac{1}{\sqrt{y}} \frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}} = \frac{1}{\Gamma(\frac{1}{2})} \left(\frac{1}{2}\right)^{\frac{1}{2}} y^{\frac{1}{2}-1} e^{-\frac{1}{2}y}$$

Note that this is the density of $\text{Gamma}(\frac{1}{2}, \frac{1}{2})$ distribution,

$$\frac{1}{\Gamma(\alpha)} \beta^\alpha y^{\alpha-1} e^{-\beta y}, \quad \text{with } \beta = \frac{1}{2}, \quad \alpha = \frac{1}{2}.$$

In the re-exprssion we used the fact that

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty x^{-1/2} e^{-x} dx = \sqrt{\pi}$$

which can be derived using variable transformation $x = y^2$.

We have shown

$$Z_1^2 \sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right)$$

Next, we derive the density function of $Z_1^2 + Z_2^2$, when Z_1, Z_2 are independent $N(0, 1)$ random variables.

Recall the additive property of Gamma random variables:

If X and Y are independent Gamma random variables with the same rate parameter, then the density of the sum is also of Gamma distribution with the same rate parameter. That is,

$$X \sim \text{Gamma}(\alpha_x, \beta), Y \sim \text{Gamma}(\alpha_y, \beta), \quad X \perp\!\!\!\perp Y \quad \implies \quad X + Y \sim \text{Gamma}(\alpha_x + \alpha_y, \beta)$$

This can be derived by density formula for the sum of two random variables (exercise).

In our derivation, since Z_1 and Z_2 are independent, consequently Z_1^2 and Z_2^2 are also independent. The additive property of Gamma distribution implies

$$Z_1^2 + Z_2^2 \sim \text{Gamma}\left(\frac{1}{2} + \frac{1}{2}, \frac{1}{2}\right) = \text{Gamma}\left(1, \frac{1}{2}\right)$$

For general n : If $Y_1, \dots, Y_n$ are independent $\sim \text{Gamma}(\alpha_i, \beta)$, using mathematical induction and the density of sum of random variables, one can prove that

$$\sum_{i=1}^n Y_i \sim \text{Gamma}\left(\sum_{i=1}^n \alpha_i, \beta\right)$$

Therefore the χ_n^2 random variable $Y = Z_1^2 + \dots + Z_n^2$ has Gamma distribution with parameters $(\frac{n}{2}, \frac{1}{2})$,

$$Z_1^2 + \dots + Z_n^2 \sim \text{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right)$$

By the density formula of Gamma distribution with $\alpha = \frac{n}{2}$, $\beta = \frac{1}{2}$,

$$f_{\chi_n^2}(y) = \frac{1}{\Gamma(\frac{n}{2})} \left(\frac{1}{2}\right)^{\frac{n}{2}} y^{\frac{n}{2}-1} e^{-\frac{1}{2}y}, \quad y \geq 0.$$

□

Mean and variance of χ^2 distribution

Since chi-square distribution is a Gamma distribution, and

$$X \sim \text{Gamma}(\alpha, \beta) \quad \Rightarrow \quad \mathbb{E}(X) = \frac{\alpha}{\beta}, \quad \text{Var}(X) = \frac{\alpha}{\beta^2}$$

we can obtain the mean and variance for χ_n^2 .

$$X \sim \chi_n^2 = \text{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right) \quad \Rightarrow \quad \mathbb{E}(X) = n, \quad \text{Var}(X) = 2n$$

An alternative, intuitive way to obtain the mean of chi-square is to use $N(0, 1)$ properties,

$$X \sim \chi_n^2, \quad \mathbb{E}(X) = \sum_{i=1}^n \mathbb{E}(Z_i^2) = n\mathbb{E}(Z_1^2),$$

and

$$Z_i \sim N(0, 1) \quad \Rightarrow \quad \mathbb{E}(Z_i^2) = \text{Var}(Z_i) = 1$$

2 The Student's t distribution

Definition of Student's t distribution

If $Z \sim N(0, 1)$, $W \sim \chi_k^2$, and Z and W are independent, then

$$T = \frac{Z}{\sqrt{W/k}} \sim t_k$$

This ratio random variable is denoted as of t distribution with k degrees of freedom.

Motivation of the t distribution

The most common test statistic for population mean parameter is of the form

$$T = \frac{\bar{X} - \mu}{\sqrt{s^2/n}}$$

where $\bar{X} = \sum_{i=1}^n X_i/n$ is the sample mean, $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is the sample variance.

We can rewrite the statistic as

$$\frac{\bar{X} - \mu}{\sqrt{s^2/n}} = \frac{\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}}{\sqrt{s^2/\sigma^2}} = \frac{Z}{\sqrt{W/(n-1)}}$$

Now it is in the form similar to the one in the definition of t random variable.

The important theorem of sampling distribution states that, if $\{X_1, \dots, X_n\}$ are *i.i.d.* $\sim N(\mu, \sigma^2)$, then

- (1) $Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$
- (2) $W = \frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$
- (3) $\bar{X}$ and s^2 are independent.

$$(4) \quad \frac{\bar{X} - \mu}{\sqrt{s^2/n}} \sim t_{n-1}$$

Part (1) follows directly from the properties of normal distribution.

The middle two steps are the core of the theorem.

Once (2) and (3) are proved, conclusion (4) follows from the definition of the t distribution.

The theorem claims that, for normal samples, the sample mean test statistic $T \sim t_{n-1}$.

For non-normal random variables, t_{n-1} is often a good approximation for small or moderate n .

For large n , t_n approach to $N(0, 1)$, consistent with the Central Limit Theorem.

Therefore this theorem of sampling distribution is very important and useful.

Density of t distribution

By the definition $T = \frac{Z}{\sqrt{W/k}} \sim t_k$, t random variable is a quotient of two random variables, thus we can derive the density function of t distribution from the density functions of normal and chi-square distributions.

- By change of variable transformation, the density of $Y = \sqrt{W/k}$ can be obtained from the density of W , which is of χ_k^2 distribution.
- The density of the ratio $T = Z/Y$ can be derived from $f_Y(y)$ — the density of Y , and $\phi(z)$ — the $N(0, 1)$ density of Z . Applying the density formula for ratio (derived in homework),

$$f_T(t) = \int_{\mathbb{R}} \phi(ty) f_Y(y) |y| dy$$

- After steps of tedious algebra derivation, the density of t_k can be obtained as

$$f_T(t) = C_k \left(1 + \frac{t^2}{k} \right)^{-\frac{k+1}{2}}$$

The constant C_k has the form

$$C_k = \frac{\Gamma(\frac{k+1}{2})}{\sqrt{k\pi} \Gamma(\frac{k}{2})} = \frac{1}{\sqrt{k} B(\frac{1}{2}, \frac{k}{2})}$$

where B is the Beta function, $B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$. In integral form, $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$.

A detailed derivation of the density function of t distribution is provided in Appendix.

Student's t and $N(0, 1)$ distribution

Both density functions of t and $N(0, 1)$ are symmetric with respect to 0, bell shaped.

The key difference is that the density of t_n has thicker tails, that is,

$$T_n \sim t_n, \quad Z \sim N(0, 1) \quad \Rightarrow \quad \mathbb{P}(|T_n| > c) > \mathbb{P}(|Z| > c) \quad \text{for large } c > 0.$$

The smaller n , the thicker the tails.

When sample size n gets large, $t_n \rightarrow N(0, 1)$ distribution as $n \rightarrow \infty$. To show the convergence limit of t_n , recall

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right) = e^x$$

Combined with continuity of exponential functions, we obtain

$$\lim_{k \rightarrow \infty} \left(1 + \frac{t^2}{k} \right)^{-\frac{k+1}{2}} = e^{-t^2/2}$$

Since the density $f_n(t)$ of $T_n \sim t_n$ has the form

$$f_n(t) \propto \left(1 + \frac{t^2}{k} \right)^{-\frac{k+1}{2}}$$

and since density functions are non-negative and integrated to 1, the dominant convergence theorem implies that the limit function must be a density function, that is,

$$\lim_{k \rightarrow \infty} f_n(t) = \lim_{k \rightarrow \infty} C_k \left(1 + \frac{t^2}{k} \right)^{-\frac{k+1}{2}} = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$$

Student's t and Cauchy distribution

When $n = 1$, the density of the t distribution becomes

$$f_1(t) = \frac{1}{\pi(1+t^2)}$$

This is also the distribution of a Cauchy random variable, a famous distribution due to its unique properties.

Apply variable substitute $x = \tan \theta$, we can derive

$$\int_{-\infty}^{\infty} \frac{1}{\pi(1+t^2)} dt = 1$$

which verifies that $f_1(t)$ indeed is a density function. However, if we try to compute the mean,

$$\int_{-\infty}^{\infty} t f_1(t) dt = \int_{-\infty}^{\infty} \frac{t}{\pi(1+t^2)} dt = +\infty$$

As $|t|$ gets larger, the tail area beyond $f(t)$ is so “fat” that the integral does not converge.

This is an important example when the Law of Large Numbers does not apply, neither the Central Limit Theorem.

3 F distribution

Another common distribution in statistical inference is the F distribution, often used in comparing nested models.

Definition of F distribution

Suppose $X \sim \chi_n^2, Y \sim \chi_m^2$, X and Y are independent. Then the quotient random variable

$$\frac{X/n}{Y/m} \sim F_{n,m}$$

is of F distribution with degrees of freedom n and m .

Density of F distribution

Using the formula for ratio random variable once more, the density of F distribution can be derived from the χ^2 distributions.

$$f_{n,m}(x) = C x^{\frac{n}{2}-1} \left(1 + \frac{n}{m}x\right)^{-\frac{n+m}{2}}$$

where the normalizing constant C can be written as

$$C = \frac{\left(\frac{n}{m}\right)^{\frac{n}{2}}}{B\left(\frac{n}{2}, \frac{m}{2}\right)} = \left(\frac{n}{m}\right)^{\frac{n}{2}} \frac{\Gamma\left(\frac{n+m}{2}\right)}{\Gamma\left(\frac{n}{2}\right)\Gamma\left(\frac{m}{2}\right)}$$

The mean of F distribution has a simple form

$$E(F) = \frac{m}{m-2}, \quad F \sim F_{n,m} \quad (m > 2)$$

which provides a good ballpark estimate for the mean, especially when the denominator degrees of freedom is large.

The derivation of the density for F distribution is provided in the Appendix.

4 The theorem of sampling distributions

In this section, we prove the most important theorem in this class, the theorem of sampling distributions. Essentially, the theorem shows that the test statistic for the normal mean is of t -distribution with $(n-1)$ degrees of freedom.

Theorem. If $X_1, \dots, X_n$ are *i.i.d.* $\sim N(\mu, \sigma^2)$, then

- (a) $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$
- (b) $\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \sim \chi_{n-1}^2$
- (c) $\bar{X}_n \perp\!\!\!\perp \sum_{i=1}^n (X_i - \bar{X}_n)^2$ ($\perp\!\!\!\perp$ means independence)
- (d) $\frac{\bar{X}_n - \mu}{\sqrt{s^2/n}} \sim t_{n-1}$

where $\bar{X}_n$ is the sample mean, $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ is the sample variance.

Proof.

- (a) For any $n > 1$, we have already derived that

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Then

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = \frac{\sqrt{n}}{\sigma} (\bar{X}_n - \mu)$$

as a linear transformation of a normal random variable is still of normal distribution, with mean

$$\mathbb{E}\left[\frac{\sqrt{n}}{\sigma} (\bar{X}_n - \mu)\right] = \frac{\sqrt{n}}{\sigma} [\mathbb{E}(\bar{X}_n) - \mu] = 0$$

and variance

$$\text{Var}\left[\frac{\sqrt{n}}{\sigma} (\bar{X}_n - \mu)\right] = \frac{n}{\sigma^2} \text{Var}(\bar{X}_n - \mu) = \frac{n}{\sigma^2} \text{Var}(\bar{X}_n) = 1$$

Therefore,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

- (b-c) We prove (b) and (c) together by mathematical induction (courtesy of Prof. S. Stigler).

First we re-express the expression

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

and show its independence of the parameter σ^2 .

Define

$$Z_i = \frac{X_i - \mu}{\sigma}, \quad i = 1, 2, \dots$$

Then for any n ,

$$\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i = \frac{\bar{X}_n - \mu}{\sigma}$$

Consequently, $X_i = \sigma Z_i + \mu$ and $\bar{X}_n = \sigma \bar{Z}_n + \mu$, and the expression

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \frac{1}{\sigma^2} \sum_{i=1}^n [(\sigma Z_i + \mu) - (\sigma \bar{Z}_n + \mu)]^2 = \sum_{i=1}^n (Z_i - \bar{Z}_n)^2$$

Therefore, to show $\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \sim \chi_{n-1}^2$ in (b) and its independence of $\bar{X}_n$ in (c), it is sufficient to show

$$\sum_{i=1}^n (Z_i - \bar{Z}_n)^2 \sim \chi_{n-1}^2$$

and

$$\sum_{i=1}^n (Z_i - \bar{Z}_n)^2 \perp\!\!\!\perp \bar{Z}_n$$

for all $n \geq 2$.

Now we prove (b) and (c) in the theorem by mathematical induction.

For $n = 2$:

$$\begin{aligned} \sum_{i=1}^2 (Z_i - \bar{Z}_2)^2 &= \left(Z_1 - \frac{Z_1 + Z_2}{2} \right)^2 + \left(Z_2 - \frac{Z_1 + Z_2}{2} \right)^2 \\ &= \left(\frac{Z_1 - Z_2}{2} \right)^2 + \left(\frac{Z_2 - Z_1}{2} \right)^2 = \left(\frac{Z_1 - Z_2}{\sqrt{2}} \right)^2 \end{aligned}$$

As a difference of two independent normal random variables, $Z_1 - Z_2$ is of normal distribution with mean and variance

$$\mathbb{E}(Z_1 - Z_2) = 0, \quad \text{Var}(Z_1 - Z_2) = \text{Var}(Z_1) + \text{Var}(Z_2) - 2\text{Cov}(Z_1, Z_2) = \text{Var}(Z_1) + \text{Var}(Z_2) = 2$$

which implies that

$$\frac{Z_1 - Z_2}{\sqrt{2}} \sim N(0, 1)$$

Consequently,

$$\left(\frac{Z_1 - Z_2}{\sqrt{2}} \right)^2 \sim \chi_1^2 = \chi_{n-1}^2$$

This proves (b) holds for $n = 2$.

To prove (c), first note that

$$\begin{bmatrix} \frac{1}{\sqrt{2}}(Z_1 - Z_2) \\ \bar{Z}_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/2 & 1/2 \end{bmatrix} \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix}$$

is of joint normal, its component covariance is

$$\begin{aligned} \text{Cov} \left(\frac{Z_1 - Z_2}{\sqrt{2}}, \bar{Z}_2 \right) &= \text{Cov} \left(\frac{Z_1 - Z_2}{\sqrt{2}}, \frac{Z_1 + Z_2}{2} \right) = \frac{1}{2\sqrt{2}} \text{Cov}(Z_1 - Z_2, Z_1 + Z_2) \\ &= \frac{1}{2\sqrt{2}} [\text{Var}(Z_1) + \text{Cov}(Z_1, Z_2) - \text{Cov}(Z_2, Z_1) - \text{Var}(Z_2)] \\ &= \frac{1}{2\sqrt{2}} [\text{Var}(Z_1) - \text{Var}(Z_2)] = 0 \end{aligned}$$

which means $\frac{Z_1 - Z_2}{\sqrt{2}}$ and $\bar{Z}_2$ are uncorrelated thus independent, by the property of normal random variables. Therefore,

$$\sum_{i=1}^2 (Z_i - \bar{Z}_2)^2 = \left(\frac{Z_1 - Z_2}{\sqrt{2}} \right)^2 \perp\!\!\!\perp \bar{Z}_2$$

which proves (c). Thus (b) and (c) in the theorem hold for $n = 2$.

For any n : Assume $\sum_{i=1}^n (Z_i - \bar{Z}_n)^2 \sim \chi_{n-1}^2$ and $\sum_{i=1}^n (Z_i - \bar{Z}_n)^2 \perp\!\!\!\perp \bar{Z}_n$.

We need to show

$$\sum_{i=1}^{n+1} (Z_i - \bar{Z}_{n+1})^2 \sim \chi_n^2 \quad \text{and} \quad \sum_{i=1}^{n+1} (Z_i - \bar{Z}_{n+1})^2 \perp\!\!\!\perp \bar{Z}_{n+1}.$$

Write

$$\begin{aligned} \sum_{i=1}^{n+1} (Z_i - \bar{Z}_{n+1})^2 &= \sum_{i=1}^n (Z_i - \bar{Z}_n + \bar{Z}_n - \bar{Z}_{n+1})^2 + (Z_{n+1} - \bar{Z}_{n+1})^2 \\ &= \sum_{i=1}^n (Z_i - \bar{Z}_n)^2 - 2 \sum_{i=1}^n (Z_i - \bar{Z}_n)(\bar{Z}_n - \bar{Z}_{n+1}) + \sum_{i=1}^n (\bar{Z}_n - \bar{Z}_{n+1})^2 + (Z_{n+1} - \bar{Z}_{n+1})^2 \end{aligned}$$

Note that the product term

$$\sum_{i=1}^n (Z_i - \bar{Z}_n)(\bar{Z}_n - \bar{Z}_{n+1}) = 0$$

we are left with

$$\begin{aligned} \sum_{i=1}^{n+1} (Z_i - \bar{Z}_{n+1})^2 &= \sum_{i=1}^n (Z_i - \bar{Z}_n)^2 + J, \\ \text{with} \quad J &= n(\bar{Z}_n - \bar{Z}_{n+1})^2 + (Z_{n+1} - \bar{Z}_{n+1})^2 \end{aligned} \tag{1}$$

To show (b) and (c), we need to prove that $J \sim \chi_1^2$ and $J \perp\!\!\!\perp \sum_{i=1}^n (Z_i - \bar{Z}_n)^2$.

Re-express the term $\bar{Z}_{n+1}$,

$$(n+1)\bar{Z}_{n+1} = n\bar{Z}_n + Z_{n+1} \quad \Rightarrow \quad \bar{Z}_{n+1} = \frac{n}{n+1}\bar{Z}_n + \frac{1}{n+1}Z_{n+1} \tag{2}$$

Now J can be expressed as

$$\begin{aligned} J &= n(\bar{Z}_n - \bar{Z}_{n+1})^2 + (Z_{n+1} - \bar{Z}_{n+1})^2 \\ &= n \left(\bar{Z}_n - \frac{n}{n+1} \bar{Z}_n - \frac{1}{n+1} Z_{n+1} \right)^2 + \left(Z_{n+1} - \frac{n}{n+1} \bar{Z}_n - \frac{1}{n+1} Z_{n+1} \right)^2 \\ &= \frac{n}{(n+1)^2} (\bar{Z}_n - Z_{n+1})^2 + \frac{n^2}{(n+1)^2} (\bar{Z}_n - Z_{n+1})^2 \\ &= \frac{n}{n+1} (\bar{Z}_n - Z_{n+1})^2 = \left(\frac{\bar{Z}_n - Z_{n+1}}{\sqrt{\frac{n+1}{n}}} \right)^2 \end{aligned}$$

As the difference of two independent normal variables, $\bar{Z}_n - Z_{n+1}$ are of normal distribution, with mean

$$\mathbb{E}(\bar{Z}_n - Z_{n+1}) = 0$$

and variance

$$\begin{aligned} \text{Var}(\bar{Z}_n - Z_{n+1}) &= \text{Var}(\bar{Z}_n) + \text{Var}(Z_{n+1}) - 2\text{Cov}(\bar{Z}_n, Z_{n+1}) \\ &= \text{Var}(\bar{Z}_n) + \text{Var}(Z_{n+1}) = \frac{1}{n} + 1 = \frac{n+1}{n} \end{aligned}$$

Thus

$$\frac{\bar{Z}_n - Z_{n+1}}{\sqrt{\frac{n+1}{n}}} \sim N(0, 1) \quad \Rightarrow \quad J = \left(\frac{\bar{Z}_n - Z_{n+1}}{\sqrt{\frac{n+1}{n}}} \right)^2 \sim \chi_1^2$$

By the induction hypothesis and the independence of the Z_i 's,

$$\bar{Z}_n \perp\!\!\!\perp \sum_{i=1}^n (Z_i - \bar{Z}_n)^2, \quad Z_{n+1} \perp\!\!\!\perp \sum_{i=1}^n (Z_i - \bar{Z}_n)^2, \quad (3)$$

Note that, since Z_{n+1} is independent of i.i.d. $Z_1, \dots, Z_n$,

$$Z_{n+1} \perp\!\!\!\perp \left[\sum_{i=1}^n \frac{\bar{Z}_n}{\sum_{i=1}^n (Z_i - \bar{Z}_n)^2} \right]$$

which means that the triple

$$\left(Z_{n+1}, \bar{Z}_n, \sum_{i=1}^n (Z_i - \bar{Z}_n)^2 \right) \quad \text{are mutually independent.} \quad (4)$$

Hence,

$$\bar{Z}_n - Z_{n+1} \perp\!\!\!\perp \sum_{i=1}^n (Z_i - \bar{Z}_n)^2$$

Consequently,

$$J = \left(\frac{\bar{Z}_n - Z_{n+1}}{\sqrt{\frac{n+1}{n}}} \right)^2 \perp\!\!\!\perp \sum_{i=1}^n (Z_i - \bar{Z}_n)^2$$

and (b) follows.

To prove (c) that $\bar{Z}_{n+1} \perp\!\!\!\perp \sum_{i=1}^{n+1} (Z_i - \bar{Z}_{n+1})^2 = \sum_{i=1}^n (Z_i - \bar{Z}_n)^2 + J$, note that (2), (3) and (4) imply

$$\bar{Z}_{n+1} \perp\!\!\!\perp \sum_{i=1}^n (Z_i - \bar{Z}_n)^2$$

In addition, $\bar{Z}_n \perp\!\!\!\perp Z_{n+1}$ yields

$$\begin{aligned} \text{Cov}(\bar{Z}_{n+1}, \bar{Z}_n - Z_{n+1}) &= \frac{1}{n+1} \text{Cov}(n\bar{Z}_n + Z_{n+1}, \bar{Z}_n - Z_{n+1}) \\ &= \frac{1}{n+1} (n\text{Var}(\bar{Z}_n) - \text{Var}(Z_{n+1})) = \frac{1}{n+1} \left(n \frac{1}{n} - 1 \right) = 0 \end{aligned}$$

Since $(\bar{Z}_{n+1}, \bar{Z}_n - Z_{n+1})$ is of bivariate normal, the uncorrelatedness means independence, hence

$$\bar{Z}_{n+1} \perp\!\!\!\perp \frac{n}{n+1} (\bar{Z}_n - Z_{n+1})^2 = J$$

By (1),

$$\bar{Z}_{n+1} \perp\!\!\!\perp \sum_{i=1}^{n+1} (Z_i - \bar{Z}_{n+1})^2 = \sum_{i=1}^n (Z_i - \bar{Z}_n)^2 + J$$

i.e., (c) follows.

By induction, we have proved that the statements (b) and (c)

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi_{n-1}^2, \quad \bar{X}_n \perp\!\!\!\perp \sum_{i=1}^n (X_i - \bar{X})^2$$

are true, for any $n > 1$.

(d) To show the statistic $\frac{\bar{X}_n - \mu}{\sqrt{s^2/n}} \sim t_{n-1}$ distribution, rewrite it as

$$\frac{\bar{X}_n - \mu}{\sqrt{s^2/n}} = \frac{\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}}{\sqrt{s^2/\sigma^2}} = \frac{Z}{\sqrt{\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 / (n-1)}}$$

By (a), the numerator $Z \in N(0, 1)$. By (b) and (c),

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \sim \chi_{n-1}^2, \quad \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \perp\!\!\!\perp \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} = Z$$

Consequently, (d) follows by the definition of t distribution. □

Appendix

This appendix contains derivation of the density of t distribution, the density of F distribution. and an alternative proof of t -distribution of the sample mean statistic,

Density function of t derivation

If

$$T = \frac{Z}{\sqrt{W/k}}, \quad Z \sim N(0, 1), \quad W \sim \chi_k^2, \quad Z \perp\!\!\!\perp W$$

then

$$T \sim t_k$$

by definition. In the following we show that T had density function

$$f_T(t) = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi} \Gamma\left(\frac{k}{2}\right)} \left(1 + \frac{t^2}{k}\right)^{-\frac{k+1}{2}}, \quad t \in \mathbb{R}$$

Proof. To derive the density function of T , use the vector variable substitution method,

$$\begin{bmatrix} T \\ V \end{bmatrix} = \begin{bmatrix} \frac{Z}{\sqrt{W/k}} \\ Z \end{bmatrix} \Rightarrow \begin{bmatrix} W \\ Z \end{bmatrix} = \begin{bmatrix} k \frac{V^2}{T^2} \\ V \end{bmatrix}$$

The joint density functions have the relation

$$f_{T,V}(t, v) |J| = f_{Z,W}(z, w), \quad J = \left| \frac{\partial(t, v)}{\partial(w, z)} \right|$$

The Jacobian

$$\left| \frac{\partial(t, v)}{\partial(w, z)} \right| = \begin{vmatrix} \frac{\partial t}{\partial w} & \frac{\partial t}{\partial z} \\ \frac{\partial v}{\partial w} & \frac{\partial v}{\partial z} \end{vmatrix} = \begin{vmatrix} z\sqrt{k}(-\frac{1}{2})w^{-3/2} & 1/\sqrt{w/k} \\ 0 & 1 \end{vmatrix} = -\frac{z\sqrt{k}}{2w^{3/2}}$$

By the independence of Z and W , their joint density function is

$$f_{Z,W}(z, w) = f_Z(z)f_W(w) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \frac{1}{2^{k/2}\Gamma(\frac{k}{2})} w^{\frac{k}{2}-1} e^{-w/2}$$

The joint density of (T, V) can be written as

$$f_{T,V}(t, v) = \frac{f_{Z,W}(z, w)}{|J|} = \frac{2}{\sqrt{2\pi} 2^{\frac{k}{2}} \Gamma(\frac{k}{2}) \sqrt{k} z} e^{-\frac{z^2}{2}} w^{\frac{k}{2}-1} e^{-\frac{w}{2}}$$

In terms of variables t and v ,

$$f_{T,V}(t, v) = \frac{1}{\sqrt{\pi} 2^{\frac{k-1}{2}} \Gamma(\frac{k}{2}) \sqrt{k} v} e^{-\frac{v^2}{2}} \left(k \frac{v^2}{t^2}\right)^{\frac{k+1}{2}} e^{-\frac{kv^2}{2t^2}} = \frac{1}{\sqrt{\pi} 2^{\frac{k-1}{2}} \Gamma(\frac{k}{2}) \sqrt{k}} \left(\frac{k}{t^2}\right)^{\frac{k+1}{2}} v^k e^{-\left(1+\frac{k}{t^2}\right)v^2/2}$$

Now we would like to derive the marginal density of T by integrating out v .

By the definition of the gamma function,

$$\int_0^\infty x^{\alpha-1} e^{-\beta x} dx = \Gamma(\alpha) \beta^{-\alpha}$$

Therefore, integration of the part in $f_{T,V}(t, v)$ containing v has the form

$$\int_0^\infty v^k e^{-\left(1+\frac{k}{t^2}\right)v^2/2} dv = \int_0^\infty 2^{\frac{k-1}{2}} x^{\frac{k+1}{2}-1} e^{-\left(1+\frac{k}{t^2}\right)x} dx = 2^{\frac{k-1}{2}} \Gamma\left(\frac{k+1}{2}\right) \left(1 + \frac{k}{t^2}\right)^{-\frac{k+1}{2}}$$

Consequently, the marginal density of T is

$$f_T(t) = \int_0^\infty f_{T,V}(t, v) dv = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{\pi} \Gamma(\frac{k}{2}) \sqrt{k}} \left(\frac{k}{t^2}\right)^{\frac{k+1}{2}} \left(1 + \frac{k}{t^2}\right)^{-\frac{k+1}{2}} = \frac{\Gamma\left(\frac{k+1}{2}\right)}{\sqrt{k\pi} \Gamma(\frac{k}{2})} \left(1 + \frac{t^2}{k}\right)^{-\frac{k+1}{2}}$$

This is the density of t_k . □

F density derivation

In the following we derive the probability density function of F distribution with degrees of freedom (n, m) .

Proof.

Consider $X = U/V$ for independent U, V , with $U \sim \chi_n^2$, $V \sim \chi_m^2$. First, we derive the density function for X .

Using the density function form for the ratio of random variables,

$$\begin{aligned} f_X(X) &= \int_0^\infty f_U(xy) f_V(y) y dy \\ &= \int_0^\infty \left(\frac{1}{\Gamma(\frac{n}{2})} \left(\frac{1}{2}\right)^{\frac{n}{2}} (xy)^{\frac{n}{2}-1} e^{-\frac{xy}{2}}\right) \left(\frac{1}{\Gamma(\frac{m}{2})} \left(\frac{1}{2}\right)^{\frac{m}{2}} y^{\frac{m}{2}-1} e^{-\frac{y}{2}}\right) y dy \\ &= x^{\frac{n}{2}-1} \frac{\left(\frac{1}{2}\right)^{\frac{n+m}{2}}}{\Gamma(\frac{n}{2})\Gamma(\frac{m}{2})} \int_0^\infty y^{\frac{n+m}{2}-1} e^{-\frac{x+1}{2}y} dy \\ &= x^{\frac{n}{2}-1} \frac{\left(\frac{1}{2}\right)^{\frac{n+m}{2}}}{\Gamma(\frac{n}{2})\Gamma(\frac{m}{2})} \Gamma\left(\frac{n+m}{2}\right) \left(\frac{2}{1+x}\right)^{\frac{n+m}{2}} \int_0^\infty \underbrace{\frac{1}{\Gamma\left(\frac{n+m}{2}\right)} \left(\frac{x+1}{2}\right)^{\frac{n+m}{2}} y^{\frac{n+m}{2}-1} e^{-\frac{x+1}{2}y}}_{\text{density of } \Gamma\left(\frac{n+m}{2}, \frac{x+1}{2}\right)} dy \end{aligned}$$

Recognizing the integrand is a Gamma density thus the integration is 1, we obtain

$$f_X(X) = \frac{1}{B\left(\frac{n}{2}, \frac{m}{2}\right)} x^{\frac{n}{2}-1} (1+x)^{-\frac{n+m}{2}}, \quad x \geq 0$$

where the beta function

$$B\left(\frac{n}{2}, \frac{m}{2}\right) = \frac{\Gamma(\frac{n}{2})\Gamma(\frac{m}{2})}{\Gamma\left(\frac{n+m}{2}\right)}$$

Next, we derive the density function for an F statistic.

The F statistic is the ratio two independent chi-square random variables, each divided by its degrees of freedom. Let

$$Y = \frac{U/n}{V/m} = \frac{m}{n} X$$

By definition,

$$Y \sim F_{n,m}$$

Y is a function of X . By the variable transformation formula, we can obtain the density of Y using the density of X just derived.

$$f_Y(y) = f_X\left(\frac{n}{m}y\right) \frac{n}{m} = \frac{1}{B\left(\frac{n}{2}, \frac{m}{2}\right)} \left(\frac{n}{m}y\right)^{\frac{n}{2}-1} \left(1 + \frac{n}{m}y\right)^{-\frac{n+m}{2}} \frac{n}{m}$$

Therefore the density of an F statistic with degrees of freedom (n, m) has the form

$$f_Y(y) = \frac{\left(\frac{n}{m}\right)^{\frac{n}{2}}}{B\left(\frac{n}{2}, \frac{m}{2}\right)} y^{\frac{n}{2}-1} \left(1 + \frac{n}{m}y\right)^{-\frac{n+m}{2}}, \quad y \geq 0.$$

□

Alternative derivation of the theorem of sampling distributions

Recall the theorem of sampling distribution can be stated as the following.

Theorem. If $X_1, \dots, X_n$ are *i.i.d.* $\sim N(\mu, \sigma^2)$, then

- (a) $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$
- (b) $\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi_{n-1}^2$
- (c) $\bar{X}_n$ and $\sum_{i=1}^n (X_i - \bar{X}_n)^2$ are independent.
- (d) $\frac{\bar{X}_n - \mu}{\sqrt{s^2/n}} \sim t_{n-1}$

To prove the theorem, the key steps are (b) and (c).

We have proved the theorem by mathematical induction, using univariate and bivariate normal properties.

In the following, we provide an alternative, more common proof of (b) and (c) using matrix algebra.

Proof.

We will prove part (c) and part (b) of the theorem by matrix algebra, using some properties of multivariate normal.

Proof of part (c)

Recall that, if random vectors V, W are independent, then their functions, say, $g(W)$ and $h(V)$, are also independent.

To prove (c), it is sufficient to show $\bar{X}$ and the sum of squares $\sum_{i=1}^n (X_i - \bar{X})^2 = (n-1)s^2$ are independent.

Use vector notation, write

$$\sum_{i=1}^n (X_i - \bar{X})^2 = U^T U, \quad U = \begin{bmatrix} X_1 - \bar{X} \\ X_2 - \bar{X} \\ \vdots \\ X_n - \bar{X} \end{bmatrix}$$

Since the sum of squares is a function of U , to prove the independence of the sum of squares and $\bar{X}$, it is sufficient to show U and $\bar{X}$ are independent. Since U is multivariate normal, it is sufficient to show that U and $\bar{X}$ are uncorrelated, which is to show that each component of U is uncorrelated with $\bar{X}$. Note that, since X_i 's are independent,

$$\text{Cov}(X_i, X_j) = 0 \quad \text{for } i \neq j.$$

Hence

$$\text{Cov}(X_i, \bar{X}) = \frac{1}{n} \text{Cov}(X_i, (X_1 + \dots + X_n)) = \frac{1}{n} \text{Cov}(X_i, X_i) = \frac{\sigma^2}{n}.$$

Consequently,

$$\text{Cov}(X_i - \bar{X}, \bar{X}) = \text{Cov}(X_i, \bar{X}) - \text{Cov}(\bar{X}, \bar{X}) = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0$$

So the components of U and $\bar{X}$ are uncorrelated. The normality implies the independence between U and $\bar{X}$.

This proves the independence of sample mean $\bar{X}$ and sample variance $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$.

Proof of (b)

We need to show that

$$\frac{1}{\sigma^2} U^T U = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2}$$

is of χ^2 distribution of $(n-1)$ degrees of freedom.

Expand the components,

$$U = \begin{bmatrix} X_1 - \bar{X} \\ X_2 - \bar{X} \\ \vdots \\ X_n - \bar{X} \end{bmatrix} = \begin{bmatrix} (1 - \frac{1}{n})X_1 - \frac{1}{n}X_2 - \dots - \frac{1}{n}X_n \\ -\frac{1}{n}X_1 + (1 - \frac{1}{n})X_2 - \dots - \frac{1}{n}X_n \\ \vdots \\ -\frac{1}{n}X_1 - \dots - \frac{1}{n}X_{n-1} + (1 - \frac{1}{n})X_n \end{bmatrix} = AX, \quad X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix}$$

The matrix A has the form

$$A = \begin{bmatrix} 1 - \frac{1}{n} & -\frac{1}{n} & \dots & -\frac{1}{n} \\ -\frac{1}{n} & 1 - \frac{1}{n} & \dots & -\frac{1}{n} \\ \vdots & \vdots & \dots & \vdots \\ -\frac{1}{n} & -\frac{1}{n} & \dots & 1 - \frac{1}{n} \end{bmatrix} = I_n - \frac{1}{n} J_n = I_n - \frac{1}{n} \begin{bmatrix} 1 & 1 & \dots & 1 \\ 1 & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \dots & 1 \end{bmatrix} [1 \ 1 \ \dots \ 1]$$

where I_n is the identity matrix of order n , and J_n is the $n \times n$ matrix with every entry = 1.

Note that, the matrix A satisfies

- $A = A^T$
- $A^2 = A$

So A is a symmetric, (orthogonal) projection matrix.

By matrix theory, A has n eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ with values $\lambda_i = 1$ or $\lambda_i = 0$ only.

Such matrix can have eigenvalue-eigenvector decomposition of the form

$$A = Q\Lambda Q^T, \quad QQ^T = Q^TQ = I_n$$

where Λ is a diagonal matrix with A 's eigenvalues $\lambda_1, \dots, \lambda_n$ on the diagonal.

Use properties of matrix trace, with is the sum of the diagonal elements of the matrix,

$$\sum_{i=1}^n a_{ii} = \text{trace}(A) = \text{trace}(Q\Lambda Q^T) = \text{trace}(\Lambda Q^TQ) = \text{trace}(\Lambda) = \sum_{i=1}^n \lambda_i$$

In the above we used the property $\text{trace}(AB) = \text{trace}(BA)$. On the other hand,

$$\sum_{i=1}^n a_{ii} = \left(1 - \frac{1}{n}\right) - \frac{1}{n} - \dots - \frac{1}{n} = n - 1$$

We conclude that there are $n - 1$ eigenvalue λ_i 's of value 1, only $\lambda_n = 0$. We now have

$$A = Q\Lambda Q^T = Q \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \\ & & & 0 \end{bmatrix} Q^T$$

and

$$U^TU = (AX)^T(AX) = (Q\Lambda Q^TX)^T(Q\Lambda Q^TX) = X^TQ\Lambda Q^TX$$

Where $Q = [q_1 \ q_2 \ \dots \ q_n]$ is the matrix of orthonormal eigenvectors of A .

Recall that A 's first $n - 1$ eigenvectors $q_1, \dots, q_{n-1}$ have eigenvalue $\lambda_j = 1$.

$$Aq_j = I_n q_j - \frac{1}{n} J_n q_j = q_j \quad \Rightarrow \quad J_n q_j = 0, \quad j = 1, \dots, n - 1.$$

Since the elements of J_n are all = 1, the sum of component elements of vector q_j must be 0. That is,

$$q_j = \begin{bmatrix} q_{1j} \\ \vdots \\ q_{nj} \end{bmatrix}, \quad \sum_{i=1}^n q_{ij} = 0 \quad \text{for } j = 1, \dots, n - 1.$$

Define an n -vector

$$Z = Q^TX = Q^T \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} = \begin{bmatrix} Z_1 \\ Z_2 \\ \vdots \\ Z_n \end{bmatrix}$$

As linear combinations of normal random variables X_i 's, the Z_i 's are also normal random variables, with

$$\mathbb{E}(Z) = Q^T \mathbb{E}(X)$$

Its first $n - 1$ components have mean 0,

$$\mathbb{E}(Z_j) = q_j^T \mathbb{E}(X) = \sum_{i=1}^n q_{ij} \mu = 0, \quad j = 1, \dots, n - 1,$$

The variance-covariance matrix of Z is

$$\text{Cov}(Z) = \text{Cov}(Q^TX) = Q^T \text{Cov}(X)Q = Q^T(\sigma^2 I_n)Q = \sigma^2(Q^TQ) = \sigma^2 I_n.$$

Thus the random variables $Z_1, \dots, Z_n$ are independent with variance σ^2 .

Recall the eigenvalues $\lambda_1 = \dots = \lambda_{n-1} = 1$, and $\lambda_n = 0$, hence

$$U^TU = X^TQ\Lambda Q^TX = (Q^TX)^T\Lambda(Q^TX) = Z^T\Lambda Z = Z_1^2 + \dots + Z_{n-1}^2$$

Note that $V_j = \frac{Z_j}{\sigma}$ ($j = 1, \dots, n - 1$) are i.i.d. $\sim N(0, 1)$, It follows that

$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} = \frac{1}{\sigma^2} U^TU = \frac{1}{\sigma^2} Z^T \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \\ & & & 0 \end{bmatrix} Z = \frac{1}{\sigma^2} (Z_1^2 + \dots + Z_{n-1}^2) = V_1^2 + \dots + V_{n-1}^2$$

and

$$V_1^2 + \dots + V_{n-1}^2 \sim \chi_{n-1}^2$$

Therefore part (b) of the theorem holds. □

We have shown the independence of $\bar{X}$ and s^2 , and $(n - 1)s^2/\sigma^2 \sim \chi_{n-1}^2$, which lead to

$$T = \frac{\bar{X} - \mu}{\sqrt{s^2/n}} \sim t_{n-1}$$

by the definition of t -distribution.